Sumit Singh Bagdwal

Software Engineering Intern — Summer 2027

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Portfolio | LinkedIn | GitHub | LeetCode

Professional Summary

Computer Science undergraduate with hands-on experience in backend development and building web applications using Node.js, Express.js, React.js, and MongoDB. Experienced in developing REST APIs, designing reliable backend systems, and writing clean, maintainable code. Strong foundation in data structures, algorithms, object-oriented programming, and software development.

Education

Graphic Era Hill University — Bhimtal, Uttarakhand
B.Tech in Computer Science and Engineering | CGPA: 8.26/10 — *July 2024 – Expected July 2028*

Technical Skills

Languages: C++, JavaScript (ES6+), Python, SQL
Backend & APIs: Node.js, Express.js, FastAPI, REST APIs, JWT, RBAC, WebSockets, Socket.IO
Databases & Cloud: MongoDB Atlas, Mongoose, MySQL, Redis, AWS EC2/S3/IAM, Docker
Frontend & Tools: React.js, Vite, Tailwind CSS, HTML/CSS, Git, GitHub Actions, Postman, Vercel
Computer Science & Practices: Data Structures & Algorithms, OOP, DBMS, Software Design, Distributed Systems, Software Development Lifecycle (SDLC), CI/CD, Agile/Scrum

Projects

Pratyaya – Payment Reliability Platform — Live Demo | GitHub
Node.js, Express.js, React.js, MongoDB Atlas

- Built a merchant-focused payment sandbox using Express.js REST APIs and React dashboards, supporting payment authorization, capture, and partial refunds across 4 demo roles.
- Implemented idempotent payment workflows, a double-entry ledger, and a transactional outbox using MongoDB ACID transactions and background jobs to maintain accurate, traceable financial records.
- Tested payment APIs and mock-provider operations using Vitest, with 30 passing API tests and 10 passing provider integration tests.

IBVAP – Intelligent Border Video Analytics — Live Demo | GitHub
Python, FastAPI, YOLO11n, AWS

- Built a border surveillance system shortlisted for Smart India Hackathon 2026, using YOLO11n and ByteTrack to detect and track objects, assess risks, and prevent duplicate alerts.
- Tested the system on 720p video using an Apple M4 CPU, achieving an average of 10.49 FPS and 91.95 ms processing latency.
- Deployed a Dockerized backend on AWS EC2, integrating FastAPI for AI processing, MySQL and Redis for data handling, and SHA-256/Ed25519 for evidence integrity verification.

Professional Experience

Lead, AWS Student Builder Group — July 2026 – Present
Student Volunteer — *Graphic Era Hill University, Bhimtal*

- Led peer learning sessions on software development and cloud computing; communicated technical concepts and helped organize the INNOVATE 1.0 campus ideathon.

Achievements & Certifications

- **Hackathons & Coding:** Smart India Hackathon 2026 shortlist; Hack the Winter 2025 national finalist; 350+ LeetCode problems solved.
- **Training:** NPTEL Data Structures and Design; AWS Cloud Essentials and Cloud Practitioner Essentials.
- **Leadership:** National Cadet Corps (NCC) B Certificate.